

EDS : First Year
Time : 2.30 hours

Subject: Oral Biology
Marks : 50

B

- Write brief relevant and legible answer.
- Illustrate your answer with well labeled diagrams wherever required.
- All questions are compulsory.

[Section B- 25 marks]

1. Describe the different anatomical aspects of Permanent Maxillary Canine? (6)
2. Enumerate the morphologic stages of tooth development. Write in detail the events during the Late Cap stage? (1+4=5)
3. Describe the development of mandible in detail. Add note on its clinical considerations? (4+1=5)
4. Write short notes on. (3x3=9)
 - a. Difference between Permanent and Primary dentition
 - b. Pharyngeal arches
 - c. Primary dentin

[Section C- 25 marks]

5. Enumerate and add note on the different histological structures of Enamel with well labeled diagram? (1+5=6)
6. Define deglutition and explain the stages of deglutition? (1+4=5)
7. Describe the contents of the dentinal tubules and add a note dentinal tubules? (2+3=5)
8. Write short notes on. (3x3=9)
 - a. Fluorosis
 - b. Dentinoenamel Junction
 - c. Dead tracts

Kathmandu University School of Medical Sciences
BDS Program
Pre-University Examination, June/July 2024

Subject: Oral Biology

Time: 2 hours 30 Minutes

Full marks: 50

Level: BDS I (New course)

Date: 30th June, 2024

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labeled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. Describe the anatomical aspects of permanent maxillary canine with well labelled diagram. [5]
2. Discuss the derivatives of pharyngeal arch. Add a note on meckels cartilage. [3+3=6]
3. What is the composition of dentin? Describe the process of dentinogenesis in detail. [1+4=5]
4. Write short notes on: [3x3=9]
 - a. Mamelons
 - b. Formation of root
 - c. Anatomy of TMJ

Section "C" [25 marks]

5. Enlist the differences between keratinocytes and nonkeratinocytes. Describe the histology of gingiva well labelled diagram. [2+4=6]
6. Enlist the stages in development of salivary glands. Differentiate between serous and mucous acini. [2+3=5]
7. Mention the functions of alveolar bone. Describe the matrix vesicle theory of mineralization. [2+3=5]
8. Write short notes on: [3x3=9]
 - a. Phases in tooth eruption
 - b. CEJ
 - c. Masticatory cycle

KATHMANDU UNIVERSITY SCHOOL OF MEDICAL SCIENCES
BDS PROGRAM
Re- Exam

Level: BDS I (new course)
Time: 1 hour 30 min

Subject: Oral Biology
Total: 30 marks

INSTRUCTIONS TO THE CANDIDATE:

- *Write brief, relevant and legible answers*
- *Illustrate your answer with well labelled diagrams wherever required*
- *All questions are compulsory*

1. Describe the anatomical aspects of permanent mandibular first molar with well labelled diagram. [5]
2. Classify oral mucosa. Describe the histology of non-keratinized epithelium with well labeled diagram. [2+3=5]
3. Enumerate hypocalcified structure of enamel. Describe the process of amelogenesis in detail. [1+4=5]
4. Write short notes on: [3x5=15]
 - a. Theories of dentinal hypersensitivity
 - b. Denticles
 - c. Bundle bone
 - d. Development of tongue
 - e. Composition and function of saliva

KATHMANDU UNIVERSITY

September/October 2023

Level : BDS

Subject: Oral Biology

Exam : BDS I

Paper : I

Time : 2 hour 30 minutes

Marks : 50

Date : September 25, 2023

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers.
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"

[25 marks]

1. Mention the composition of enamel. Discuss Amelogenesis in detail with diagrams. [1+5=6]
2. Describe in detail the morphology of permanent maxillary first molar with diagram. [5]
3. Describe the embryological development of palate. [5]
4. Write short notes on: [3×3=9]
 - a. HERS
 - b. Anatomy of TMJ
 - c. Endochondral ossification

SECTION "C"

[25 marks]

5. Discuss the functions of pulp. Enumerate the histological zones of pulp. [2+3=5]
6. Describe in detail about histology, functional significance and age changes of Maxillary sinus. [2+2+2=6]
7. Enlist the tongue papilla present in human. Add a note on any two of them with well labelled diagram. [1+2+2=5]
8. Write short notes on: [3×3=9]
 - a. Ductal modification of saliva.
 - b. Theories of dentinal hypersensitivity.
 - c. Neonatal line.

Kathmandu University School of Medical Sciences
BDS Program
Pre-University Examination, August 2023

Subject: Oral Biology

Time: 2 hours & 30 Minutes

Full marks: 50

Level: BDS I (New course)

Date: 8th August 2023

INSTRUCTIONS TO THE CANDIDATE

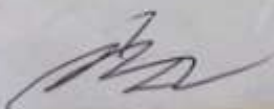
- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labeled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. Describe the development of mandible along with its clinical consideration. Add a note on Meckel's cartilage. [3+2=5]
2. Describe in detail the morphology of permanent mandibular first molar. [5]
3. Enlist the fate of dental lamina. Add a note on Amelogenesis along with its clinical consideration. [2+4=6]
4. Write short notes on: [3x3=9]
 - a. Collagen synthesis
 - b. Differences between primary and permanent dentition
 - c. Alveolar bone proper

Section "C" [25 marks]

5. Differentiate between keratinocytes and nonkeratinocytes. Describe the histology of gingiva. [2+4=6]
6. What is the composition of saliva? Add a note on various ducts of salivary glands. [2+3=5]
7. Discuss the age changes in dentin. Differentiate between mantle and circumpulpal dentin. [3+2=5]
8. Write short notes on: [3x3=9]
 - a. Taste pathway
 - b. Phases of eruption
 - c. Theories of mineralization



KATHMANDU UNIVERSITY
December 2022/January 2023

Level : BDS

Subject: Oral Biology

Exam : BDS I

Paper : I

Time : 2 hour 30 minutes

Marks : 50

Date : January 10, 2023

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers.
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"

[25 marks]

1. Write the chronology of permanent dentition. Discuss the difference in the occlusal morphology of maxillary and mandibular first molar. [2+4=6]
2. Discuss in detail the prenatal development of tongue with its clinical considerations. [4+1=5]
3. Enumerate the hypocalcified structures of enamel. Add a note on any two of them. [1+2+2=5]
4. Write short notes on: [3×3=9]
 - a. Formation and secretion of saliva
 - b. Functions of pulp
 - c. Tubercle and cusp

SECTION "C"

[25 marks]

5. Classify cementum. Differentiate between cellular and acellular cementum with well labeled diagram. [2+4=6]
6. Classify oral mucosa. Add a note on non-keratinocytes. [1+4=5]
7. Define alveolar process. Enumerate the functions of alveolar bone and discuss about bundle bone. [1+2+2=5]
8. Write short notes on: [3×3=9]
 - a. Stages of deglutition
 - b. Modulation cycle
 - c. Dead tracts

Kathmandu University School of Medical Sciences
BDS Program
Pre-University Examination, July 2021

Subject: Oral Biology

Time: 2 hours 30 Minutes

Full marks: 50

Level: BDS Ist (New course)

Date: 25th July 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labeled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. Enumerate the transient structures during tooth development. Add a note on theories of tooth patterning. [1+4=5]
2. Describe the anatomical aspects of permanent maxillary left first molar with well labelled diagram. [6]
3. Describe the life cycle of ameloblasts. [5]
4. Write short notes on: [3x3=9]
 - a. Classification of malocclusion
 - b. Development of tongue and palate
 - c. Theories of mineralization

Section "C" [25 marks]

5. Classify oral mucosa. Describe various taste buds and taste pathway. [2+3=5]
6. Describe about development of salivary glands. Add a note on ductal modification of saliva. [3+3=6]
7. Define alveolar process. Explain the structure of alveolar bone. [1+4=5]
8. Write short notes on: [3x3=9]
 - a. Zones of pulp
 - b. Cemental repair
 - c. Tertiary dentin

Kathmandu University School of Medical Sciences
BDS Program
Pre-University Examination, July 2021

Subject: Oral Biology

Time: 2 hours 30 Minutes

Full marks: 50

Level: BDS Ist (New course)

Date: 25th July 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labeled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. Enumerate the transient structures during tooth development. Add a note on theories of tooth patterning. [1+4=5]
2. Describe the anatomical aspects of permanent maxillary left first molar with well labelled diagram. [6]
3. Describe the life cycle of ameloblasts. [5]
4. Write short notes on: [3x3=9]
 - a. Classification of malocclusion
 - b. Development of tongue and palate
 - c. Theories of mineralization

Section "C" [25 marks]

5. Classify oral mucosa. Describe various taste buds and taste pathway. [2+3=5]
6. Describe about development of salivary glands. Add a note on ductal modification of saliva. [3+3=6]
7. Define alveolar process. Explain the structure of alveolar bone. [1+4=5]
8. Write short notes on: [3x3=9]
 - a. Zones of pulp
 - b. Cemental repair
 - c. Tertiary dentin

Kathmandu University School of Medical Sciences
BDS Program
Internal Examination, April 2021

Subject: Oral Biology

Time: 2 hours & 30 Minutes

Full marks: 50

Level: BDS I (New course)

Date: 9th April 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labeled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. Describe anatomy of Temporomandibular joint. Add a note on its histology. [2+3=5]
2. Differentiate between maxillary and mandibular first premolar. [5]
3. Explain the terms differentiation, maturation and turnover rate in oral epithelium. Differentiate between keratinized and nonkeratinized epithelium. [2+4=6]
4. Write short notes on: [3 x 3 = 9]
 - a. Basket cells
 - b. Functions of lymphatic system
 - c. Antrum of Highmore

Section "C" [25marks]

5. Describe the physiological stages of tooth formation. [5]
6. Enumerate the theories of eruption. Add a note on the most accepted theory of eruption. [2+4=6]
7. Enumerate hypocalcified structures of enamel. Describe the age changes in oral soft tissue. [1+4=5]
8. Write short notes on: [3 x 3 = 9]
 - a. Incremental growth lines in enamel vs dentin vs cementum
 - b. Neural crest cells
 - c. Factors regulating cementogenesis

Kathmandu University School of Medical Sciences
BDS Program
Internal Examination, April 2021

Subject: Oral Biology

Time: 2 hours & 30 Minutes

Full marks: 50

Level: BDS I (New course)

Date: 9th April 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labeled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. Describe anatomy of Temporomandibular joint. Add a note on its histology. [2+3=5]
2. Differentiate between maxillary and mandibular first premolar. [5]
3. Explain the terms differentiation, maturation and turnover rate in oral epithelium. Differentiate between keratinized and nonkeratinized epithelium. [2+4=6]
4. Write short notes on: [3 x 3 = 9]
 - a. Basket cells
 - b. Functions of lymphatic system
 - c. Antrum of Highmore

Section "C" [25marks]

5. Describe the physiological stages of tooth formation. [5]
6. Enumerate the theories of eruption. Add a note on the most accepted theory of eruption. [2+4=6]
7. Enumerate hypocalcified structures of enamel. Describe the age changes in oral soft tissue. [1+4=5]
8. Write short notes on: [3 x 3 = 9]
 - a. Incremental growth lines in enamel vs dentin vs cementum
 - b. Neural crest cells
 - c. Factors regulating cementogenesis

KATHMANDU UNIVERSITY
March/April 2021

Level : BDS

Subject: Oral Biology

Exam : BDS I

Paper :

Time : 2 hr. 30 minutes

Marks : 50

Date : April 7, 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Enumerate the different morphological stages of development of tooth. Describe the nutritional supply and changes happening during each stage in detail. Add a note on dental lamina and its fate. [1+3+2=6]
2. Describe amelogenesis. Add a note on physical and chemical properties of enamel. [3+1+1=5]
3. Enumerate various elevated anatomical landmarks of teeth. Describe ridges and explain different types of ridges present in the posterior teeth. [2+3=5]
4. Write short notes on: [3×3=9]
 - a. Development of tongue and its clinical considerations
 - b. Various tooth numbering system
 - c. HER's and its clinical significance

SECTION "C"
[25 marks]

5. Classify oral mucous membrane. Describe the histology of taste buds. Add a note on taste mapping. [2+3+1=6]
6. Describe various theories of tooth eruption. [5]
7. Classify salivary glands. Describe the histology of mucous salivary gland. [2+3=5]
8. Write short notes on: [3×3=9]
 - a. Shedding of teeth
 - b. Enamel lamellae, enamel tuft and enamel spindle
 - c. Histology of temporomandibular joint

Kathmandu University School of Medical Sciences
BDS Program
Internal Examination, April 2021

Subject: Oral Biology

Time: 2 hours & 30 Minutes

Full marks: 50

Level: BDS I (New course)

Date: 9th April 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labeled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. Describe anatomy of Temporomandibular joint. Add a note on its histology. [2+3=5]
2. Differentiate between maxillary and mandibular first premolar. [5]
3. Explain the terms differentiation, maturation and turnover rate in oral epithelium. Differentiate between keratinized and nonkeratinized epithelium. [2+4=6]
4. Write short notes on: [3 x 3 = 9]
 - a. Basket cells
 - b. Functions of lymphatic system
 - c. Antrum of Highmore

Section "C" [25marks]

5. Describe the physiological stages of tooth formation. [5]
6. Enumerate the theories of eruption. Add a note on the most accepted theory of eruption. [2+4=6]
7. Enumerate hypocalcified structures of enamel. Describe the age changes in oral soft tissue. [1+4=5]
8. Write short notes on: [3 x 3 = 9]
 - a. Incremental growth lines in enamel vs dentin vs cementum
 - b. Neural crest cells
 - c. Factors regulating cementogenesis

Kathmandu University School of Medical Sciences
BDS Program
Internal Examination, April 2021

Subject: Oral Biology

Time: 2 hours & 30 Minutes

Full marks: 50

Level: BDS I (New course)

Date: 9th April 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labeled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. Describe anatomy of Temporomandibular joint. Add a note on its histology. [2+3=5]
2. Differentiate between maxillary and mandibular first premolar. [5]
3. Explain the terms differentiation, maturation and turnover rate in oral epithelium. Differentiate between keratinized and nonkeratinized epithelium. [2+4=6]
4. Write short notes on: [3 x 3 = 9]
 - a. Basket cells
 - b. Functions of lymphatic system
 - c. Antrum of Highmore

Section "C" [25marks]

5. Describe the physiological stages of tooth formation. [5]
6. Enumerate the theories of eruption. Add a note on the most accepted theory of eruption. [2+4=6]
7. Enumerate hypocalcified structures of enamel. Describe the age changes in oral soft tissue. [1+4=5]
8. Write short notes on: [3 x 3 = 9]
 - a. Incremental growth lines in enamel vs dentin vs cementum
 - b. Neural crest cells
 - c. Factors regulating cementogenesis

Kathmandu University School of Medical Sciences
BDS Program
Pre-University Examination, November 2022

Subject: Oral Biology

Time: 2 hours 30 Minutes

Full marks: 50

Level: BDS I (New course)

Date: 25th November 2022

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labeled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. Enlist the fate of dental lamina. Describe early bell stage in detail. [2+3=5]
2. Define crest of curvature. Describe the occlusal aspect of mandibular first molar with well labeled diagram. [1+4=5]
3. How does TMJ differ from other synovial joint? Describe the anatomy of TMJ. [2+4=6]
4. Write short notes on: [3x3=9]
 - a. Surface structures of enamel
 - b. Development and nerve supply of tongue
 - c. Theories of dentinal hypersensitivity

Section "C" [25 marks]

5. Enlist the differences between bone and cementum. Add a note on Cementogenesis. [2+3=5]
6. Enumerate the nonkeratinocytes present in oral mucosa. Describe in detail about any two of them. [1+2+2=5]
7. What is the composition of bone? List the functions of alveolar bone. Explain about bundle bone. [1+2+3=6]
8. Write short notes on: [3x3=9]
 - a. Denticles
 - b. Intermediate plexus
 - c. Phases of deglutition

Kathmandu University School of Medical Sciences
BDS Program
Internal Examination, June 2022

Subject: Oral Biology

Time: 2 hours 30 Minutes

Full marks: 50

Level: BDS I (New course)

Date: 6th June, 2022

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labeled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. What is the composition of enamel? Discuss Amelogenesis in detail. [1+5=6]
2. Describe in detail the morphology of permanent maxillary left canine with well labeled diagram. [5]
3. Discuss the functions of pulp. Add a note on Plexus of Raschkow. [3+2=5]
4. Write short notes on: [3x3=9]
 - a. Line angles and point angles
 - b. Hypocalcified structures of enamel
 - c. Elevated landmarks of tooth

Section "C" [25 marks]

5. Enumerate the morphological stages of tooth development. Discuss about transient structures. [1+4=5]
6. Classify salivary glands. Differentiate between serous and mucous acini. [2+3=5]
7. Enlist the layers of dental apparatus. Add a note on reciprocal induction and describe the process of formation of root. [1+2+3=6]
8. Write short notes on: [3x3=9]
 - a. Tooth numbering system
 - b. Occlusal aspect of maxillary first premolar
 - c. Formation and secretion of saliva

KATHMANDU UNIVERSITY
University Examination
February/March 2022

Level : BDS

Subject: Oral Biology

Exam : BDS I

Paper :

Time : 2 hour 30 minutes

Marks : 50

Date : March 20, 2022

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Describe in detail the morphology of permanent maxillary right canine with well labeled diagram. 3 [6]
2. Classify dentin. Differentiate between primary and secondary dentin. 1.5 [2+3=5] 3
3. List the developmental stages of odontogenesis. Add a note on the different cell layers noted during early bell stage. 3 [1+4=5]
4. Write short notes on: [3×3=9]
 - a. FDI clinical notation 1.5
 - b. Meckel's cartilage 1.5
 - c. Neonatal line 1

SECTION "C" [25 marks]

5. Enumerate various salivary glands. Add a note on major salivary glands in detail. [1+5=6] 3
6. Classify oral mucosa. Describe the histological features of normal gingiva. [1+4=5] 3
7. Define tooth eruption. Explain the stages of tooth eruption. [1+4=5] 3
8. Write short notes on: [3×3=9]
 - a. Defense cells of pulp
 - b. Taste buds
 - c. Reciprocal induction in development of tooth

KATHMANDU UNIVERSITY SCHOOL OF MEDICAL SCIENCES
BDS PROGRAM
Pre-university Examination- 2019

Level: BDS 1st year (new course)
Time: 2 hours 30 minutes

Subject: Oral Biology

INSTRUCTIONS TO THE CANDIDATE:

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labelled diagrams wherever required
- All questions are compulsory

Section B [25 marks]

1. Describe occlusal morphology of permanent mandibular second premolar. Differentiate between primary and permanent dentition. [3+3=6]
2. Enumerate the physiological stages of tooth development. Discuss Amelogenesis in detail. [1+4=5]
3. Describe the composition and functions of saliva. Add a note on secretory cells. [1+2+2=5]
4. Write short notes on: [3 x 3 = 9]
 - a) Pharyngeal phase of deglutition
 - b) Curve of spee
 - c) Cemental repair

Section C [25 marks]

5. Classify oral mucosa. Describe the histological features of keratinized mucosa in detail. [1+4=5]
6. Discuss functions of Periodontal Ligament. Add a note on its Principal fibers. [1+5=6]
7. Enumerate the functions of alveolar bone. Describe alveolar bone proper in detail. [2+3=5]
8. Write short notes on: [3 x 3 = 9]
 - a. Matrix vesicle theory
 - b. Korff's fiber
 - c. Incremental lines in enamel vs Incremental lines in cementum

Kathmandu University School of Medical Sciences
BDS Program
Internal Examination, May 2019

Subject: Oral Biology
Time: 2 hours 30 Minutes
Full marks: 50

Level: BDS I (New course)
Date: 19th May, 2019

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labelled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. Discuss the development of tongue. Add a note on blood supply and nerve supply of it. [4+2=6]
2. Describe the anatomical aspects of permanent mandibular central incisor with well labelled diagram. [5]
3. Enumerate the theories of dentinal hypersensitivity. Describe the most accepted theory in detail. [1+4=5]
4. Write short notes on: [3 x 3 = 9]
 - a) Desmosomes
 - b) Primary curvature vs secondary curvature
 - c) Ultrastructure and course of enamel rods

Section "C" [25 marks]

5. Describe the life cycle of ameloblasts with a well labelled diagram. Add a note on modulation cycle of ameloblasts. [4+2=6]
6. Discuss the arch traits of incisors. [5]
7. Enumerate the parts of the dental apparatus. Describe the histo-differentiation stage of tooth development. [1+4=5]
8. Write short notes on: [3 x 3 = 9]
 - a) Lobe vs Mamelons
 - b) Tooth numbering system
 - c) Tome's process vs granular layer of tomes

Kathmandu University School of Medical Sciences

BDS Program

Pre University Examination, December 2018

Subject: Oral Biology

Time: 2 hours & 30 Minutes

Full marks: 50

Level: BDS I (New course)

Date: 2nd December, 2018

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labeled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. Describe occlusal morphology of permanent maxillary first molar. Differentiate between maxillary and mandibular first premolar. [3+3=6]
2. Classify cementum. Discuss Cementogenesis in detail. [1+4=5]
3. Describe formation and secretion of saliva. Add a note on ductal modification of saliva. [3+2=5]
4. Write short notes on: [3x3=9]
 - a) Cell junctions
 - b) Development of roots of teeth
 - c) Landmarks of tooth

Section "C" [25 marks]

5. Classify oral mucosa. Describe the histological features of gingiva. [1+4=5]
6. Enlist various stages of amelogenesis. Explain hypocalcified structures of enamel. [1+5=6]
7. Enumerate the theories of tooth eruption. Discuss the ligament traction theory in detail. [1+4=5]
8. Write short notes on: [3x3=9]
 - a) Angle's classification of malocclusion
 - b) Histology of TMJ
 - c) Primary vs Secondary vs Tertiary dentin

Kathmandu University School of Medical Sciences
BDS Program
Internal Examination, April 2023

Subject: Oral Biology
Time: 2 hours 30 minutes
Full marks: 50

Level: BDS I (New course)
Date: 28th April 2023

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labelled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. Enlist four genes involved in odontogenesis. Describe the theories of tooth patterning. [2+4=6]
2. Classify dentin. Differentiate between primary, secondary and tertiary dentin. [1+4=5]
3. Describe in detail the morphology of permanent maxillary right central incisor with well labeled diagram. [5]
4. Write short notes on: [3x3=9]
 - a. Tooth numbering system
 - b. Occlusal aspect of maxillary first premolar
 - c. Neural crest cells

Section "C" [25 marks]

5. Enlist the surface structures of enamel. Describe in detail about enamel rods. Add a note on imbrication lines of Pickerill. [1+3+2=6]
6. What is the composition of cementum? Describe about various types of Cementoenamel junction. [1+4=5]
7. Discuss the functions of periodontal ligament. Add a note on its synthetic cells. [2+3=5]
8. Write short notes on: [3x3=9]
 - a. Line angles and point angles
 - b. Cemental repair
 - c. Tomes' process

Kathmandu University School of Medical Sciences

BDS Program

Pre University Examination, December 2018

Subject: Oral Biology

Time: 2 hours & 30 Minutes

Full marks: 50

Level: BDS I (New course)

Date: 2nd December, 2018

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answers with well labeled diagrams wherever required
- All questions are compulsory

Section "B" [25 marks]

1. Describe occlusal morphology of permanent maxillary first molar. Differentiate between maxillary and mandibular first premolar. [3+3=6]
2. Classify cementum. Discuss Cementogenesis in detail. [1+4=5]
3. Describe formation and secretion of saliva. Add a note on ductal modification of saliva. [3+2=5]
4. Write short notes on: [3x3=9]
 - a) Cell junctions
 - b) Development of roots of teeth
 - c) Landmarks of tooth

Section "C" [25 marks]

5. Classify oral mucosa. Describe the histological features of gingiva. [1+4=5]
6. Enlist various stages of amelogenesis. Explain hypocalcified structures of enamel. [1+5=6]
7. Enumerate the theories of tooth eruption. Discuss the ligament traction theory in detail. [1+4=5]
8. Write short notes on: [3x3= 9]
 - a) Angle's classification of malocclusion
 - b) Histology of TMJ
 - c) Primary vs Secondary vs Tertiary dentin

Kathmandu University School of Medical Sciences
Department of Dental
BDS 1st Year (New Course)
1st Internal Examination, 2018

Subject: **Oral Biology**

Time: 2 hours 30 minutes

Full marks: 50

Date: 23rd April, 2018

INSTRUCTIONS TO THE CANDIDATE:

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labelled diagrams wherever required
- All questions are compulsory

Section B [25marks]

1. Enumerate physiological stages of tooth development. Describe advanced bell stage. [2+4=6]
2. Describe the anatomical aspects of permanent maxillary canine with well labelled diagram. [5]
- ✓ 3. Enlist the type of human dentition. Differentiate between deciduous and permanent teeth. [1+4=5]
4. Write short notes on: [3 x 3 = 9]
 - a) Fate of dental lamina
 - b) Transient structures
 - c) Lobe vs cingulum vs mamelon

Section C [25marks]

5. Enumerate the layers of dental apparatus. Describe life cycle of ameloblasts. [1+4=5]
- ✓ 6. Describe development of tongue. Add a note on innervation and clinical consideration. [3+1+1=5]
- ✓ 7. Describe development of palate and lips. Add a note on neural crest cells. [3+3=6]
8. Write short notes on: [3 x 3 = 9]
 - ✓ a) Pharyngeal arches and pouch
 - b) Tooth numbering system
 - c) Theories of tooth patterning

*****Good Luck*****

Dr. Pratibha

KATHMANDU UNIVERSITY
May/June 2017

Level : BDS

Subject: Oral Biology

Exam : BDS I (new course)

Paper :

Time : 2 hour 30 minutes

Marks : 50

Date : June 21, 2017

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Explain in detail the development of tongue. [5]
2. What are the different muscles of mastication? Explain about temporalis muscle. [2+4=6]
3. Enumerate the different physiologic stages of tooth development. Write in detail about the events during the bell stage of tooth development. [1+4=5]
4. Write short notes on: [3×3=9]
 - a. Neural crest cells
 - b. Dentinogenesis
 - c. Mamelons

SECTION "C" [25 marks]

5. What are zones of the pulp? Explain the function of the pulp. [1+5=6]
6. Write the composition of saliva. Explain in detail the formation and pathway of secretion of saliva. [1+4=5]
7. Describe the histology of cementum. [5]
8. Write short notes on: [3×3=9]
 - a. Sclerotic dentin
 - b. Production of voice
 - c. Cementoenamel junction

23

KATHMANDU UNIVERSITY
December 2016/January 2017

Level : BDS

Subject: Oral Biology

Exam : BDS I (new course)

Paper :

Time : 2 hour 30 minutes

Marks : 50

Date : January 12, 2017

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Enumerate the derivatives of the Pharyngeal Arch System. Write briefly on the development of the palate. [3+3=6]
2. Describe the anatomy and nerve supply of parotid gland. Write the composition of saliva. [4+1=5]
3. Describe the embryological development of palate. [5]
4. Write short notes on: [3 × 3=9]
 - a. Intramembranous bone formation
 - b. Difference between primary and permanent dentition
 - c. Secretory stage of Amelogenesis

SECTION "C" [25 marks]

5. Classify dentin. Describe the age and functional changes in dentin. [2+3=5]
6. List the components of the temporomandibular joint. Describe briefly the biomechanics involved in Mastication. [1+4=5]
7. Describe the zones of pulp with a neat labelled diagram. Add a note on the functions of the pulp. [4+2=6]
8. Short notes: [3 × 3=9]
 - a. Types of cements/enamel junction
 - b. Dead tracts
 - c. Stages of deglutition

Level: B25
Exam: BDS I (new course)
Time: 2 hr, 30 minutes
Date: June 7, 2016

Subject: Oral Surgery
Paper:
Marks: 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Classify tooth traits. Discuss the class traits of incisors. Add a note on variations seen in incisors. [2+3+1=6]
2. Enumerate the morphological and histological stages of tooth development. Describe in detail changes seen during early bell stage of tooth development. Add a note on the fate of dental lamina. [1+2+1=5]
3. Write down the steps in dentine formation. Add a note on dentine hypersensitivity. [3+2=5]
4. Write short notes on: [3×3=9]
 - a. Mamelons
 - b. Meckel's cartilage
 - c. Successional lamina

SECTION "C" [25 marks]

5. Classify salivary glands. Discuss the histology of major salivary gland. Describe the ultra-structural characteristics of mucous cell. [2+2+2=6]
6. Define periodontium. Enumerate the principal fibers of periodontal ligament. Describe the histology of gingival sulcus. [1+2+2=5]
7. Discuss calcium – phosphate metabolism. [5]
8. Write short notes on: [3×3=9]
 - a. Cement-enamel junctions
 - b. Muscles of mastication
 - c. Pulp stones

Level: B25
Exam: BDS I (new course)
Time: 2 hr, 30 minutes
Date: June 7, 2016

Subject: Oral Surgery
Paper:
Marks: 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Classify tooth traits. Discuss the class traits of incisors. Add a note on variations seen in incisors. [2+3+1=6]
2. Enumerate the morphological and histological stages of tooth development. Describe in detail changes seen during early bell stage of tooth development. Add a note on the fate of dental lamina. [1+2+1=5]
3. Write down the steps in dentine formation. Add a note on dentine hypersensitivity. [3+2=5]
4. Write short notes on: [3×3=9]
 - a. Mamelons
 - b. Meckel's cartilage
 - c. Successional lamina

SECTION "C" [25 marks]

5. Classify salivary glands. Discuss the histology of major salivary gland. Describe the ultra-structural characteristics of mucous cell. [2+2+2=6]
6. Define periodontium. Enumerate the principal fibers of periodontal ligament. Describe the histology of gingival sulcus. [1+2+2=5]
7. Discuss calcium – phosphate metabolism. [5]
8. Write short notes on: [3×3=9]
 - a. Cement-enamel junctions
 - b. Muscles of mastication
 - c. Pulp stones

KATHMANDU UNIVERSITY
2016

Level : BDS

Subject: Oral biology

Exam : BDS I

Paper : 1

Time : 3 hour

Marks : 70

Date : November 8, 2016

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "A" [35 marks]

1. Describe the anatomical aspects of Permanent Maxillary 1ST Premolar. [5]
2. Define Mastication. Describe in detail the different phases of Masticatory cycle. [1+4=5]
3. Enlist the types of human dentition. Discuss the difference between Permanent dentition and deciduous dentition. [1+4=5]
4. Define Mineralization. Enumerate the various theories of Mineralization and discuss Matrix Vesicle theory in detail. [1+1+3=5]
5. Write short notes on: [3×5=15]
 - a. Occlusal aspect of Maxillary 1st Molar
 - b. Composition of Saliva
 - c. Tooth numbering system
 - d. Ridges and grooves

SECTION "B" [35 marks]

6. Enumerate the Physiological stages of Tooth development. Describe the events during the late Cap stage. [1+4=5]
7. Enlist the various types of Dentine. Describe Primary Dentin in detail. [2+3=5]
8. Classify Oral Mucosa. Discuss in detail about lining Mucosa. [1+4=5]
9. Enlist the developmental processes responsible for development of face. Discuss development of tongue in detail. [1+4=5]
10. Write short notes on: [3×5=15]
 - a. Alveolar bone proper
 - b. Odontoblasts Vs Ameloblasts
 - c. Zones of pulp
 - d. Serous Acini
 - e. Mechanism of Eruptive tooth movement

KATHMANDU UNIVERSITY
July/August 2018

Level : BDS

Subject: Oral Biology

Exam : BDS I (new course)

Paper :

Time : 2 hour 30 minutes

Marks : 50

Date : July 20, 2018

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Describe the occlusal morphology of permanent mandibular first molar. Differentiate between permanent maxillary and mandibular 1st molar. [3+3=6]
2. Enumerate the various developmental stages of tooth. Write about the advanced bell stage of tooth development with labeled diagram. [2+3=5]
3. Write down the stages in life cycle of ameloblasts. What is the fate of ameloblast? [4+1=5]
4. Write short notes on: [3×3=9]
 - a. Embryological development of tongue
 - b. Occlusal aspect of maxillary first premolar
 - c. Hertwig's epithelial root sheath

SECTION "C" [25 marks]

5. What is periodontium? Describe the principal fibers of periodontal ligament with diagram. Mention the functions of periodontal ligament. [1+3+2=6]
6. Classify salivary gland. Mention the composition and functions of saliva. [2+3=5]
7. Classify dentin. Explain the theories of dentinal hypersensitivity. [2+3=5]
8. Write short notes on: [3×3=9]
 - a. Classification of cementum
 - b. Neonatal line
 - c. Tooth eruption theories

Level : BDS

Subject: Oral Surgery

Exam : BDS I

Paper : I

Time : 3 hrs.

Marks : 70

Date : June 7, 2016

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "A" [35 marks]

1. Describe the morphological features of permanent maxillary canine with well labelled diagrams. [5]
2. Define occlusion. Add a note on Angle's classification of malocclusion. [1+4=5]
3. Describe the structure of a taste bud with well labelled diagram. Add a note on the taste pathway from the anterior 2-3rd of the tongue. [3+2=5]
4. Differentiate between deciduous and permanent dentition. [5]
5. Write short notes on: [5×3=15]
 - a. Pharyngeal phase of deglutition
 - b. Functions of saliva
 - c. Tooth numbering systems
 - d. Masseter
 - e. Production of voice

SECTION "B" [35 marks]

6. List the derivatives of the first brachial arch. Write briefly the development of the palate. [2+3=5]
7. Classify dentin. Write in detail about reparative dentin. [2+3=5]
8. Describe the theories of tooth eruption. Add a note on the role of Gubernacular chord in the eruption of permanent teeth. [4+1=5]
9. List the major salivary glands. Describe the histological features of the mixed acini. Add a note on the importance of myoepithelial cells. [1+2+2=5]
10. Write short notes on: [5×3=15]
 - a. Histological feature of normal keratinized mucosa
 - b. Cemento - enamel junction
 - c. Functions of maxillary sinus
 - d. Dentinal hypersensitivity
 - e. HERS

KATHMANDU UNIVERSITY
2016

Level : BDS

Subject: Oral Biology

Exam : BDS I (new course)

Paper :

Time : 2 hr. 30 minutes

Marks : 50

Date : November 8, 2016

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Write down the different types of Dentin. Describe briefly the process of Dentinogenesis. [2+3=5]
2. Describe the differences between permanent maxillary first and second premolar. Describe the different types of occlusal aspect of mandibular second premolar. [4+2=6]
3. Describe in detail the various developmental stages of tooth development. [5]
4. Short notes: [3×3= 9]
 - a. Advanced bell stage
 - b. Mantle dentin
 - c. Pulp stones

SECTION "C" [25 marks]

5. List down the differences between cellular and acellular cementum. Describe briefly the functions of the periodontal ligament. [3+3=6]
6. Classify salivary glands. Describe the structure of terminal secretory units. [1+4=5]
7. Define oral mucosa. Enumerate the cell of oral mucosa. What are the functions of oral mucosa. [1+1+3=5]
8. Write short notes on: [3×3=9]
 - a. Myoepithelial cells.
 - b. Theories of tooth eruption
 - c. Physiology of taste

20731/3/16

Level : BDS

Subject: Oral biology

Exam : BDS I

Paper : 1

Time : 3 hour

Marks : 70

Date : November 8, 2016

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "A" [35 marks]

1. Describe the anatomical aspects of Permanent Maxillary 1st Premolar. [5]
2. Define Mastication. Describe in detail the different phases of Masticatory cycle. [1+4=5]
3. Enlist the types of human dentition. Discuss the difference between Permanent dentition and deciduous dentition. [1+4=5]
4. Define Mineralization. Enumerate the various theories of Mineralization and discuss Matrix Vesicle theory in detail. [1+1+3=5]
5. Write short notes on: [3×5=15]
 - a. Occlusal aspect of Maxillary 1st Molar
 - b. Composition of Saliva
 - c. Tooth numbering system
 - d. Ridges and grooves

SECTION "B" [35 marks]

6. Enumerate the Physiological stages of Tooth development. Describe the events during the late Cap stage. [1+4=5]
7. Enlist the various types of Dentine. Describe Primary Dentin in detail. [2+3=5]
8. Classify Oral Mucosa. Discuss in detail about lining Mucosa. [1+4=5]
9. Enlist the developmental processes responsible for development of face. Discuss development of tongue in detail. [1+4=5]
10. Write short notes on: [3×5=15]
 - a. Medial bone growth
 - b. Developmental stages of tooth
 - c. Dental arches
 - d. Dental arches

KATHMANDU UNIVERSITY

200...

Level: BDS

Exam: BDS I

Subject: Oral Biology

Paper: I

Time: 3 hrs

Marks: 70

Date:

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams wherever required
- All questions are compulsory

Section A

20 Marks

Oral physiology & Oral Embryology

1. Describe the composition of saliva. Discuss in detail the functions of saliva. Add a note on clinical considerations of saliva. (2+2+1=5)
2. Define deglutition. Write in detail on stages of deglutition. (1+4=5)
3. Write Short Notes on: (2X5=10)
 - a. Cap stage
 - b. Speech
 - c. GCF (Gingival crevicular fluid)
 - d. Mastication
 - e. Physiology of taste

Section B

25 Marks

Dental Anatomy (Morphology)

4. Describe in detail the occlusal surface of maxillary first molar. Differentiate between maxillary and mandibular first molar. (2+3=5)
5. Differentiate between deciduous dentition and permanent dentition. Enumerate the functions of deciduous dentition. (3+2=5)
6. Define occlusion. Describe different types of mandibular movements. (1+4=5)
7. Short Notes: (2X5=10)
 - a. Memmelons
 - b. Tooth numbering system
 - c. oblique ridge
 - d. embrassure
 - e. occlusal view of mandibular second premolar

Section C

25 Marks

Oral Histology

8. Describe physical and chemical properties of enamel. Write in detail on enamel rod. (1+1+3=5)
9. Define pulp. Describe different zones of pulp. (2+3=5)
10. Enumerate the groups of principle fibers of periodontal ligament. Add a note on functions of PDL. (2+3=5)
11. Short notes (2x5=10)
 - a. Enamel spindle
 - b. Striae of retzius
 - c. CEJ
 - d. Cellular cementum vs. acellular cementum
 - e. Dead track

KATHMANDU UNIVERSITY

200...

Level: BDS

Exam: BDS I

Subject: Oral Biology

Paper: I

Time: 3 hrs

Marks: 70

Date:

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams wherever required
- All questions are compulsory

Section A

20 Marks

Oral physiology & Oral Embryology

1. Describe the composition of saliva. Discuss in detail the functions of saliva. Add a note on clinical considerations of saliva. (2+2+1=5)
2. Define deglutition. Write in detail on stages of deglutition. (1+4=5)
3. Write Short Notes on: (2X5=10)
 - a. Cap stage
 - b. Speech
 - c. GCF (Gingival crevicular fluid)
 - d. Mastication
 - e. Physiology of taste

Section B

25 Marks

Dental Anatomy (Morphology)

4. Describe in detail the occlusal surface of maxillary first molar. Differentiate between maxillary and mandibular first molar. (2+3=5)
5. Differentiate between deciduous dentition and permanent dentition. Enumerate the functions of deciduous dentition. (3+2=5)
6. Define occlusion. Describe different types of mandibular movements. (1+4=5)
7. Short Notes: (2X5=10)
 - a. Memmelons
 - b. Tooth numbering system
 - c. oblique ridge
 - d. embrassure
 - e. occlusal view of mandibular second premolar

Section C

25 Marks

Oral Histology

8. Describe physical and chemical properties of enamel. Write in detail on enamel rod. (1+1+3=5)
9. Define pulp. Describe different zones of pulp. (2+3=5)
10. Enumerate the groups of principle fibers of periodontal ligament. Add a note on functions of PDL. (2+3=5)
11. Short notes (2x5=10)
 - a. Enamel spindle
 - b. Striae of retzius
 - c. CEJ
 - d. Cellular cementum vs. acellular cementum
 - e. Dead track

Dr. Pratiksha

KATHMANDU UNIVERSITY
2014

Level : BDS

Subject: Oral Biology

Exam : BDS I

Paper : I

Time : 3 hrs.

Marks : 70

Date : September 14, 2014

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "A" [35 marks]

1. Describe the occlusal surface of permanent mandibular first molar with a neat labelled diagram. Differentiate between permanent mandibular and maxillary first molar. [3+2=5]
2. Differentiate between deciduous and permanent teeth. Enumerate the functions of deciduous teeth. [3+2=5]
3. Define deglutition. Write in detail the stages of deglutition and its neural control. [1+4=5]
4. Classify the muscles of mastication. Discuss the origin, insertion, relation, innervation and blood supply of masseter muscle. [1+4=5]
5. Write short notes on: [5×3=15]
 - a. Tooth numbering system
 - b. Functions of saliva
 - c. Differentiate between ridge and tubercles
 - d. Physiology of taste
 - e. Metabolism of calcium

SECTION "B" [35 marks]

6. Describe the composition and physical properties of enamel. Add a note on surface structures of enamel. [3+2=5]
7. Describe the zones of pulp with a neat labeled diagram. Add a note on its clinical consideration. [4+1=5]
8. Enumerate the junctions in oral mucosa. Describe dentogingival junction. [1+4=5]
9. Describe in detail the morphological stages of development of tooth. Draw a neat labelled diagram of cap stage. [3+2=5]
10. Write short notes on: [5×3=15]
 - a. Histology of maxillary sinus
 - b. Cytoskeletal junctions
 - c. Enamel lamellae vs enamel crack
 - d. Nonkeratinocytes
 - e. Neural crest cells

KATHMANDU UNIVERSITY
2013

Level : BDS

Subject: Oral Biology

Exam : BDS I

Paper :

Time : 3 hrs.

Marks : 70

Date : September 17, 2013

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "A" [35 marks]

1. Describe the morphology of permanent mandibular first molar in detail. Draw a well labelled diagram of its occlusal surface. [3+4=7]
2. Describe the Histological structure of full developed dental organ. Give the function and fate of its constituent parts. [3+4=7]
3. Describe the status of dentition at the age of eight years. [5]
4. Describe the various stages in the life cycle of ameloblast. [3+3=6]
5. Write short notes on: [5×2=10]
 - a. Mantle dentine
 - b. Lamina dura
 - c. Howship lacunae
 - d. Ridge
 - e. Embrasure

SECTION "B" [35 marks]

6. Describe the histological structure of enamel. Add a note on cemento enamel junction. [3+4=7]
7. Describe the principal fibre of the periodontal membrane. Draw a well labelled diagram. [4+3=7]
8. Describe the main histological structure of dentine. [6]
9. Describe the age changes in pulp and dentine. [5]
10. Write short notes on : [5×2=10]
 - a. Bundle bone
 - b. Sharpey's fibre
 - c. Korr's fibres
 - d. Cingulum
 - e. Fossa

KATHMANDU UNIVERSITY
2012

Level : BDS
Exam : BDS I
Time : 3 hrs.

Subject: Oral Biology
Paper : I
Marks : 70

Date : May 16, 2013

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "A" [35 marks]

1. Describe the different stages of Dentition. Describe the function of Deciduous teeth. [2+3=5]
2. Mention the chronology of Eruption of Permanent teeth. Describe the dentition status of a nine year old child. [3+4=7]
3. Describe the anatomical and physiological features of Deciduous dentition. Explain in brief the anatomy of Deciduous maxillary central incisor. [3+4=7]
4. Describe the development of mandible in detail with diagrams. [3+3=6]
5. Write short notes on: [5×2=10]
 - a) Deglutition
 - b) Curve of spee
 - c) Embrasures
 - d) Bundle bone and lamina dura
 - e) Line angel & point angel

SECTION "B" [35 marks]

6. Classify oral mucus membrane. Draw a well labeled diagram of the oral epithelium. Enumerate the differences between Masticatory and Lining mucosa. [3+4=7]
7. Describe the various Papillae of Tongue. Write a note on taste buds. [2+3=5]
8. Describe the anatomical and histological structure of Pulp with well labeled diagram and its function. [3+4=7]
9. Write the differences between $4 \times 5 = 6$
 $(1+2=3)$
 - a) Acellular and ~~Ac~~ellular cementum
 - b) Enamel tufts & enamel spindle
 - c) Dentino enamel junction and cemento enamel junction
 - d) Granular layer of Tomes and Contour lines of Owen.
10. Write short notes on: [5×2=10]
 - a) Sharpey's fibres
 - b) Enamel rods
 - c) Epithelial attachment of gingival
 - d) Transparent dentine
 - e) Inter globular dentine

KATHMANDU UNIVERSITY
2012

Level : BDS
Exam : BDS I

Subject : Oral Biology
Paper :

Time : 3 hrs.

Full Marks : 70

Date : September 16, 2012

INSTRUCTION TO THE CANDIDATE

- Answer each section in a separate answer book.
- Write brief, relevant and legible answers.
- Illustrate your answer with well labeled diagrams wherever required.
- All questions are compulsory.

SECTION "A" [35 marks]

1. Describe occlusal morphology of mandibular first permanent molar with labeled diagram.
Add a note on radicular anatomy. [3+1+1=5]
2. Define mastication. Explain stages of mastication. Mention the factors affecting mastication. [1+2+2=5]
3. Enumerate the difference between deciduous and permanent dentition, in terms of: [2+2+1=5]
 - a. Morphology
 - b. Pulpal anatomy
 - c. Histology
4. Define physiologic occlusion. Differentiate between centric relation and centric occlusion.
Discuss the importance of normal occlusion. [1+2+2=5]
5. Write short notes on: [3×5=15]
 - a. Composition of saliva
 - b. FDI tooth numbering system
 - c. Cusp of Carabelli
 - d. Resonance of speech
 - e. Morphology of maxillary first premolar

SECTION "B" [35 marks]

6. Describe the histologic structure of enamel. Explain the stages of amelogenesis.
Add a note on dentino-enamel junction. [2+2+1=5]
7. Define periodontium. Describe the biochemical composition of cementum.
Classify cementum. [2+2+1=5]
8. Enumerate developmental processes responsible for the development of palate. Describe the embryological sequence in the development of secondary palate with diagram. [1+3+1=5]
9. Define oral mucosa. Enumerate various cells of oral mucosa. Describe the functions of oral mucosa. [2+1+2=5]
10. Write short notes on: [3×5=15]
 - a. Anatomy of temporomandibular joint
 - b. Dental lamina
 - c. Theories of tooth eruption
 - d. Meckel's cartilage
 - e. Histology of parotid gland

KATHMANDU UNIVERSITY
2011

Level : BDS
Exam : BDS I

Subject : Oral Biology
Paper :

Time : 3 hrs.

Marks : 70

Date : September 5, 2011

INSTRUCTION TO THE CANDIDATE

- Answer each section in a separate answer book.
- Write brief, relevant and legible answers.
- Illustrate your answer with well labeled diagrams wherever required.
- All questions are compulsory.

SECTION "A" [35 marks]

1. Describe morphology of permanent mandibular first molar in detail. Draw a well labeled diagram of its occlusal surface. [4+2=6]
2. Differentiate between deciduous dentition and permanent dentition, in terms of; [2+2+2=6]
 - a) Morphology
 - b) Chronology
 - c) Histology
3. Define occlusion. Differentiate between centric occlusion and centric relation. Add a note on lateral occlusion. [1+3+2=6]
4. Define mastication. Describe phases of mastication. Explain the factors affecting mastication. [2+3+2=7]
5. Write short notes on: [5×2=10]
 - a) Composition of saliva
 - b) Theories of mineralization of tooth
 - c) Resonance in speech
 - d) Ridges on teeth
 - e) Palmer-Zsigmondy tooth numbering system

SECTION "B" [35 marks]

6. Describe the histological structure of enamel. Explain Striae of Retzius and Bands of Hunter and Schreger. [3+2+2=7]
7. Define oral mucosa. Enumerate its various types. Explain functions of oral mucosa. [1+2+3=6]
8. Discuss embryological development of tooth. Explain various stages of tooth development. Add a note on dental follicle. [2+3+1=6]
9. Describe embryological development of mandible. Enumerate embryological processes responsible for development of mandible. Mention the fate of Meckel's cartilage. [3+2+1=6]
10. Write short notes on: [5×2=10]
 - a) Maxillary sinus
 - b) Histological structure of alveolar bone
 - c) Gubernacular canal
 - d) Cement, -enamel junction
 - e) Functions of pulp

Dr. Pratibha

KATHMANDU UNIVERSITY
2009

Level : BDS
Exam : BDS I

Subject : Oral Biology
Paper :

Time : 3 hrs.

Marks : 70

Date : February 5, 2010

INSTRUCTION TO THE CANDIDATE

- Answer each section in a separate answer book.
- Write brief, relevant and legible answers.
- Illustrate your answer with well labeled diagrams wherever required.
- All questions are compulsory.

SECTION "A" [20 marks]

1. Describe the embryological development of mandible. Add a note on Meckle's cartilage. [3+2=5]
2. Enlist and explain various theories on eruption of tooth. Differentiate between emergence and eruption of tooth. [2+2+1=5]
3. Define mastication. Describe various phases of mastication. [2+3=5]
4. Explain mechanism of calcium and phosphate deposition of teeth. Discuss effects of various hormones affecting calcium and phosphate metabolism. [3+2=5]

SECTION "B" [25 marks]

5. Enumerate various elevated anatomical landmarks of teeth. Explain different types of ridges present in posterior teeth. [2+3=5]
6. Describe occlusal surface of permanent upper first molars with labeled diagram. Explain normal occlusal relationship of upper and lower first molars. [3+2=5]
7. Enlist the periods of development of dentition. Discuss pre-dentition period in detail. [2+3=5]
8. Enlist and describe normal spaces present in between the teeth in children. Explain 'Leeway space of Nance'. [2+2+1=5]
9. Discuss general anatomical features of premolars. Differentiate between the morphology of upper first and lower first premolars. [2+3=5]

SECTION "C" [25 marks]

10. Explain enamel rod with well labeled diagram. Discuss physical and chemical properties of enamel. [2+1+2=5]
11. Describe histological zones of pulp with diagram. Enlist functions of pulp. [3+2=5]
12. Enumerate cells of the periodontal ligament. Describe the functions of various cells of periodontal ligament. [2+3=5]
13. Enlist major salivary glands. Discuss functions of saliva. [2+3=5]

(2)

KATHMANDU UNIVERSITY
2010

Level : BDS
Exam : BDS I

Subject : Oral Biology
Paper :

Time : 3 hrs.

Marks : 70

Date : August 9, 2010

INSTRUCTION TO THE CANDIDATE

- Answer each section in a separate answer book.
- Write brief, relevant and legible answers.
- Illustrate your answer with well labeled diagrams wherever required.
- All questions are compulsory.

SECTION "A" [20 marks]

1. Enumerate developmental processes responsible for development of lower third of the face. Describe the embryological development of tongue. [2+3=5]
2. Explain the mechanism of physiological exfoliation of human dentition. Add a note on gubernacular canal. [4+2=6]
3. Write short notes on: [3×3=9]
 - a. Fluoride metabolism
 - b. Deglutition
 - c. Articulators of speech

SECTION "B" [25 marks]

4. Describe the morphology of coronal portion of maxillary right central incisor. Draw a well labeled diagram of lingual surface of maxillary central incisor. [3+2=5]
5. Discuss the general anatomical features of mandibular first molar. Describe the morphology of occlusal surface of mandibular first molar with diagram. [2+3+1=6]
6. Define occlusion. Differentiate between centric occlusion and centric relation. [2+3=5]
7. Write short notes on: [3×3=9]
 - a. Gum pad
 - b. Pit and fissures
 - c. Anthropoid space

SECTION "C" [25 marks]

8. Describe the histological features of dentin with its types. Draw a well labeled diagram of odontoblastic process. [2+2+2=6]
9. Explain and classify cemento-enamel junction. Add a note on gingival margin.

10.

KATHMANDU UNIVERSITY
2008

Level : MBBS
Exam : BDS I

Subject : Oral Biology
Paper : *Handwritten signature*

Time : 3 hrs.

Marks : 70

Date : December 7, 2008

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INSTRUCTION TO THE CANDIDATE

- Answer each section in a separate answer book.
- Write brief, relevant and legible answers.
- Illustrate your answer with well labeled diagrams wherever required.
- All questions are compulsory.

SECTION "A" [20 marks]

1. Describe embryological development of palate. Enumerate the reasons of change in position of palatal selves during the development of palate. [3+2=5]
2. Enumerate theories on eruption of a tooth. Discuss in detail about vascular pressure theory. [2+3=5]
3. Write short notes on: [5+2=10]
 - a. Infantile swallowing
 - b. Bud stage
 - c. Function of saliva
 - d. Taste buds
 - e. Retained deciduous teeth

SECTION "B" [25 marks]

1. Enumerate the morphologic, physiologic and chronologic differences between permanent and deciduous dentition. [2+2+1=5]
2. Describe anatomy of lower first permanent molar. Differentiate the difference between upper and lower first permanent molar. [2+3=5]
3. Enumerate various tooth numbering system and discuss the most preferred system in clinical practice and mention its advantage. [2+2+1=5]
4. Write short notes on: [5+2=7]
 - a. Cingulum
 - b. Endodontic anatomy of

Dr. Prabha

SECTION "C" [25 marks]

8. Enumerate the layers of oral mucosa. Describe the structure of oral mucosa in different region of the oral cavity. [2+3=5]
9. What is the odontogenic region of pulp? Draw a well labelled diagram and write in detail about odontoblast. [1+2+2=5]
10. Enumerate all the structures of enamel and write in detail on enamel rod. [2+3=5]
11. Write short notes on: [5×2=10]
 - a. Enamel tufts
 - b. Incremental lines of Retzius
 - c. Dentinoenamel junction
 - d. Cellular cementum vs. acellular cementum
 - e. Sharpey's fibres

Pre University 2021 (set B)

Section B [25marks]

1. Enumerate the morphological stages of tooth development. Add a note on Bell stage. [1+4=5]
2. Describe the anatomical aspects of permanent mandibular right first molar with well labelled diagram. [6]
3. Describe the hypocalcified structures of enamel. [5]
4. Write short notes on: [3 x 3 = 9]
 - a) Fiber system of Periodontal Ligament.
 - b) Neural crest cells
 - c) Types of ossification.

Section C [25marks]

5. Classify oral mucosa. Describe histology of parakeratinized mucosa. [2+3=5]
6. Describe functions of salivary glands. Add a note on ductoacinar unit with diagram. [3+3=6]
7. Define alveolar process. Enumerate the functions of alveolar bone and add a note on bundle bone. [1+2+2=5]
8. Write short notes on: [3 x 3 = 9]
 - a. Cells of pulp
 - b. Incremental lines of Salter
 - c. Shedding of tooth

BDS: First Year
Time: 2:30hrs

Subject: Oral Biology
Marks: 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book.
- Write brief, relevant and legible answer.
- Illustrate your answer with well labeled diagrams wherever required.
- All questions are compulsory.

Section B (25 marks)

1. Discuss the occlusal aspect of Permanent Mandibular 1st Molar. Write difference between Maxillary 1st Premolar and Mandibular 1st premolar? (3+3=6)
2. Define Dentinogenesis? Discuss the process of Dentinogenesis in detail? (1+4=5)
3. Describe the anatomical aspect of Permanent Maxillary Molar with Diagram? (4+1=5)
4. Write short notes on: (3x3=9)
 - a. Tooth numbering systems
 - b. Development of Dentino gingival junction
 - c. Embrasures and Fissures

Section C (25 marks)

5. Write in detail the Composition and function of Saliva. Discuss the role of Striated duct in salivary modification? (2+2+2=6)
6. Classify Oral mucosa. Write in brief on the specialized mucosa? (1+4=5)
7. Describe the histology of pulp in detail with a neat labeled diagram? (4+1=5)
8. Write short notes on: (3x3=9)
 - a. Cemento enamel junctions
 - b. Alveolar bone proper
 - c. Sharpey's fibers

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PAGE

Dr. Pratibha

BDS : First Year
Time : 2.30 hours

Subject: Oral Biology
Marks : 50

- > Write brief relevant and legible answer.
- > Illustrate your answer with well labeled diagrams wherever required.
- > All questions are compulsory.

[Section B- 25 marks]

1. Describe the different anatomical aspects of Permanent Maxillary Canine? (6)
2. Enumerate the morphologic stages of tooth development. Write in detail the events during the Late Cap stage? (1+4=5)
3. Describe the development of mandible in detail. Add note on its clinical considerations? (4+1=5)
4. Write short notes on. (3x3=9)
 - a. Difference between Permanent and Primary dentition
 - b. Pharyngeal arches
 - c. Primary dentin

[Section C- 25 marks]

5. Enumerate and add note on the different histological structures of Enamel with well labeled diagram? (1+5=6)
6. Define deglutition and explain the stages of deglutition? (1+4=5)
7. Describe the contents of the dentinal tubules and add a note dentinal tubules? (2+3=5)
8. Write short notes on. (3x3=9)
 - a. Fluorosis
 - b. Dentinoenamel Junction
 - c. Dead tracts

INSTRUCTION TO THE CANDIDATE

Dr. Pratibha Patel

- Answer each section in a separate answer book.
- Write brief, relevant and legible answers.
- Illustrate your answer with well labeled diagrams wherever required.
- All questions are compulsory.

SECTION "A" [20 marks]

1. Describe in detail the different stages of tooth development. [5]
2. Describe the various age changes in teeth and jaws. [2+3=5]
3. Write short notes on: [5×2=10]
 - a. Mastication
 - b. Shedding of deciduous teeth
 - c. Reduced enamel epithelium
 - d. Stensen's duct
 - e. Chemical composition of saliva

SECTION "B" [25 marks]

4. Describe a contact point. Describe in detail the occlusal anatomy of permanent maxillary molar of left side. [2+2+1=5]
5. What are the features common to all canines? Name the differences between maxillary canine and mandibular canine. [2+3=5]
6. What is the shape of enamel rod and how it is positional in the crown of the tooth? [2+2+1=5]
7. Write short notes on: [5×2=10]
 - a. Mamelon
 - b. Groove
 - c. Axial wall
 - d. Tubercle
 - e. Nerve supply of tongue